SUNWAY UNIVERSITY

Jeffrey Cheah Foundation
Nurturing the Seeds of Wisdom

Practical 03: Setting up a HTCondor Cluster

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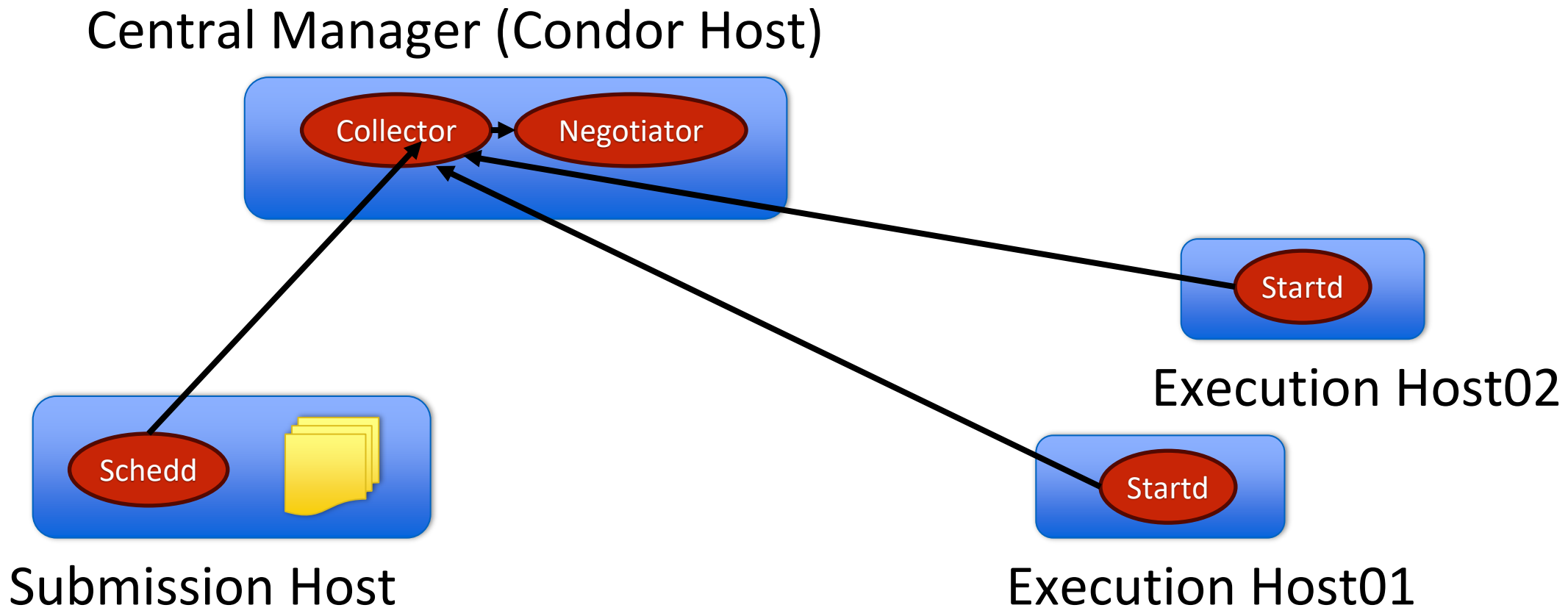
NET3204
Distributed Systems

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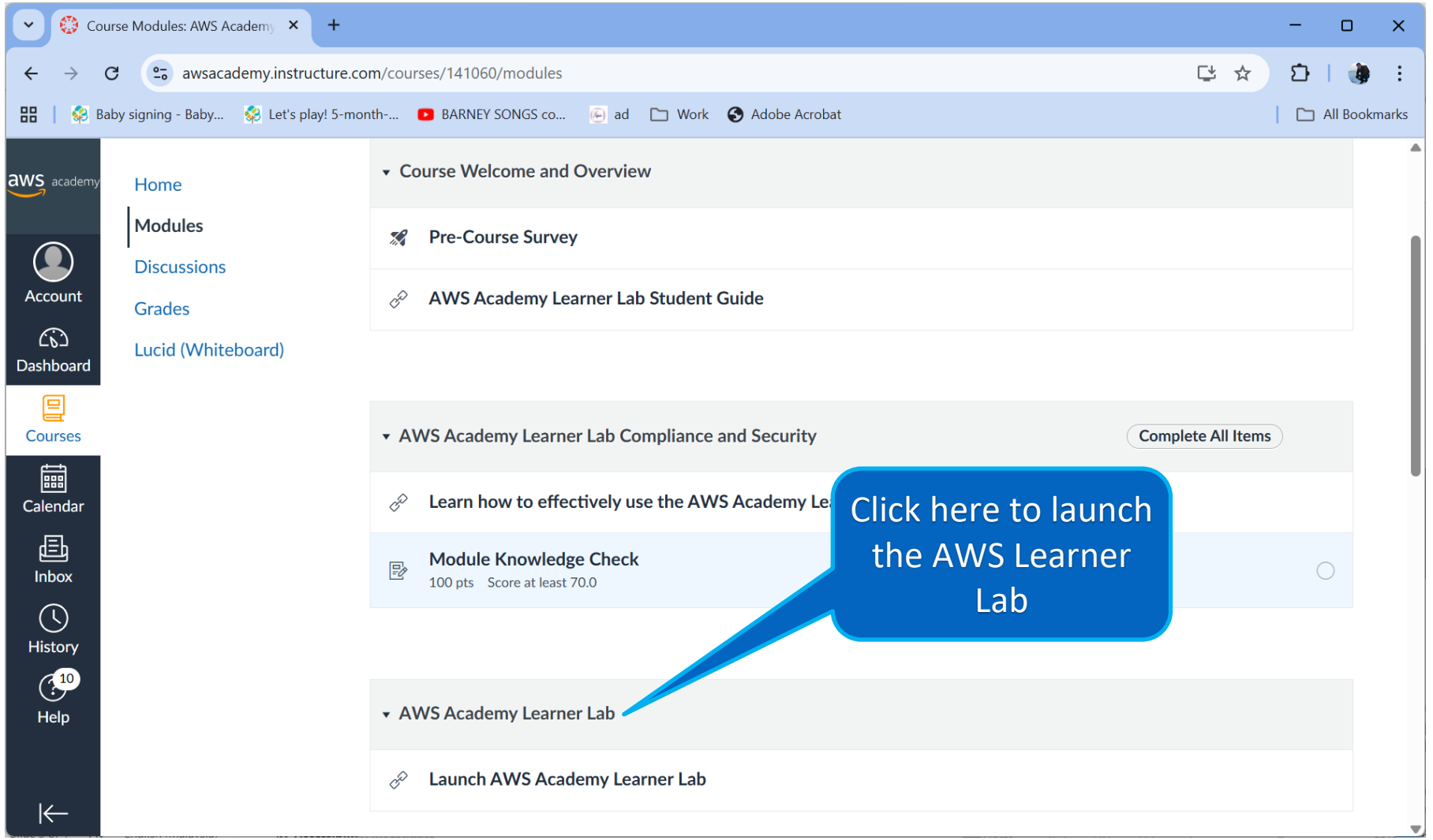
Practical Description

- In this practical, you will learn how to:
  - Sign in to the AWS Academy
  - Launch FOUR virtual machine from the AWS EC2
  - Set up a HTCondor pool with a Central Manager, Submission Host, and 2 Execution Hosts
  - Submit a sample job to the pool

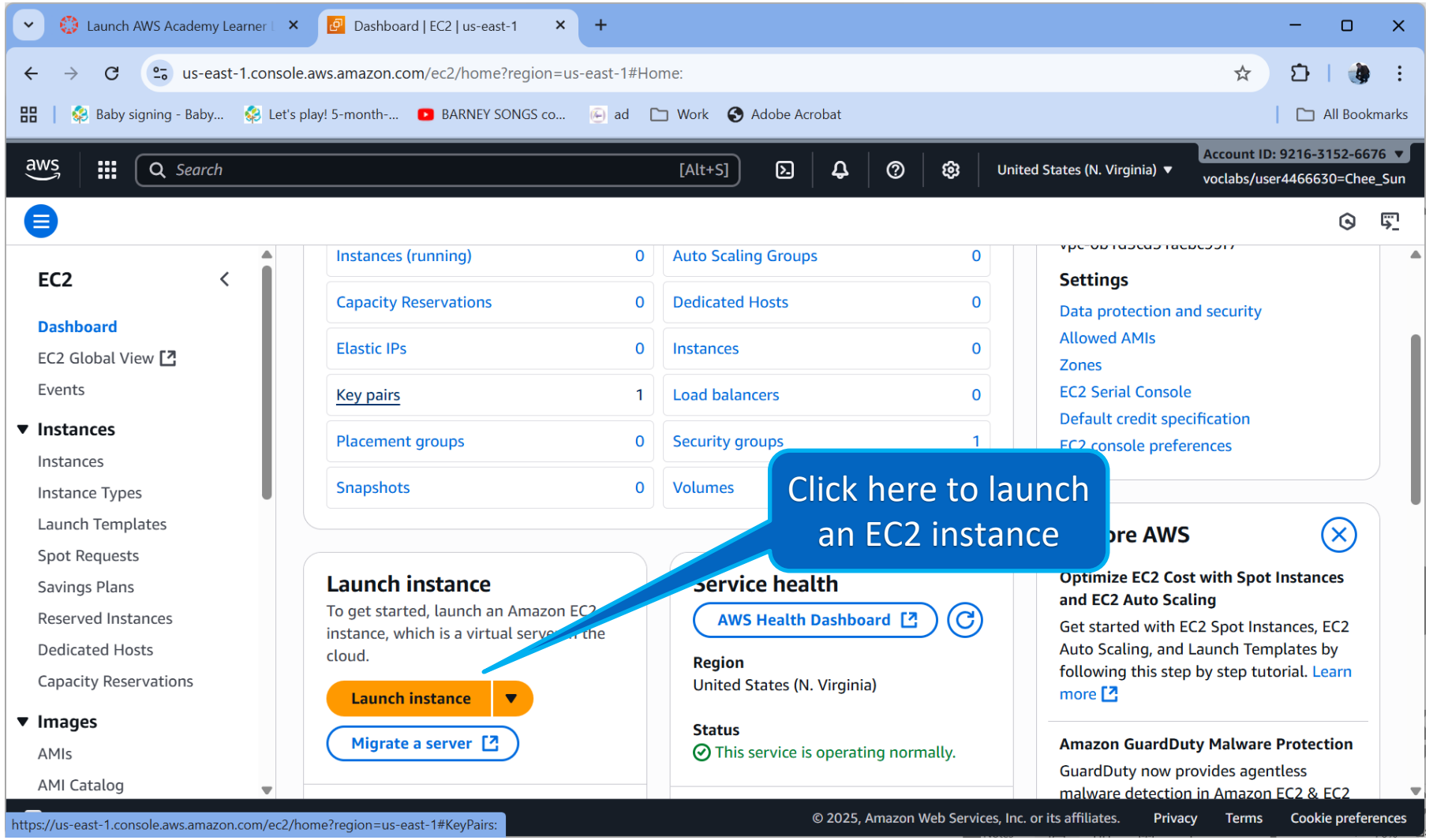
Our deployment plan today



1. Sign in to the Course Page

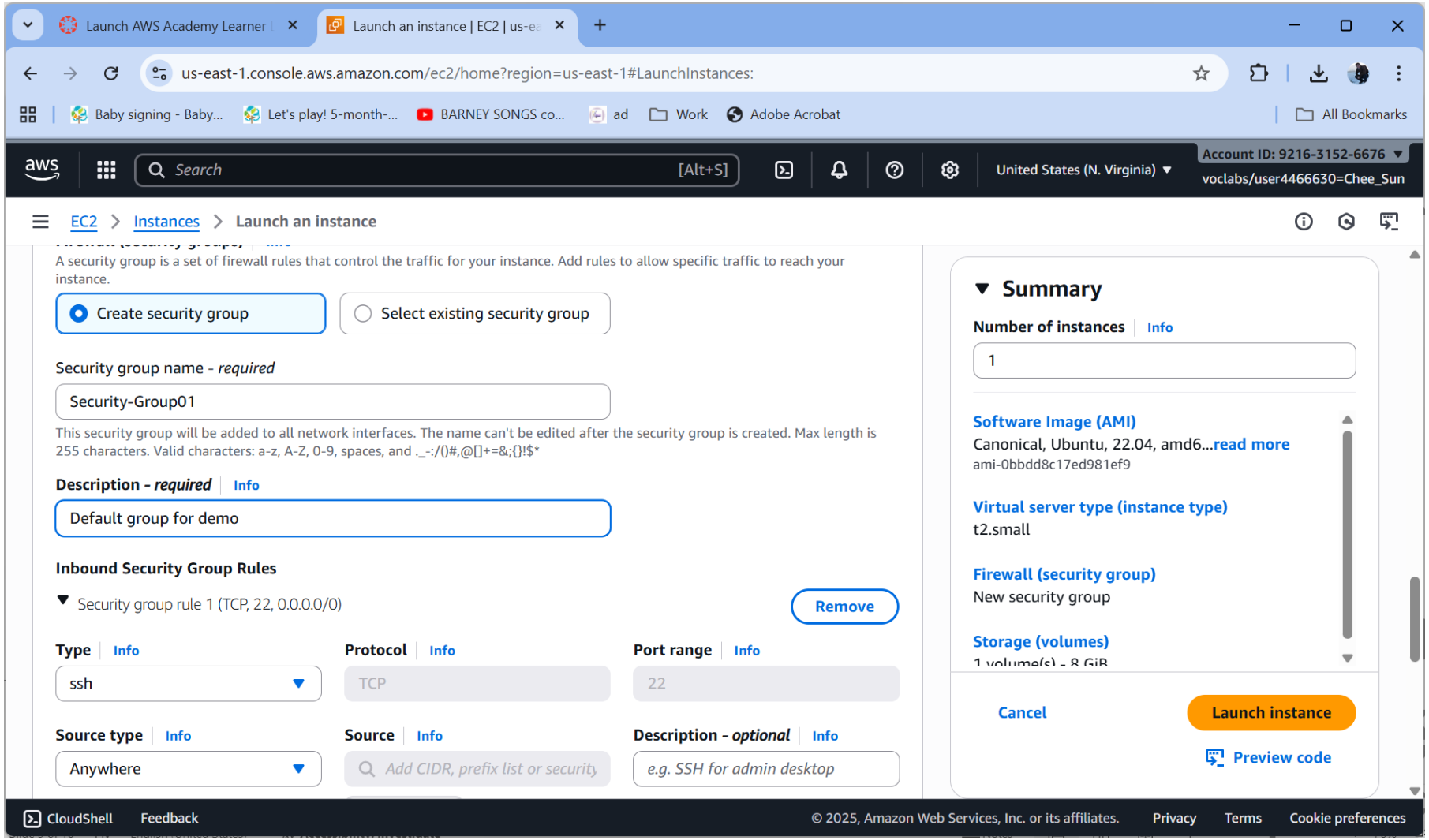


2. Launch FOUR EC2 instance



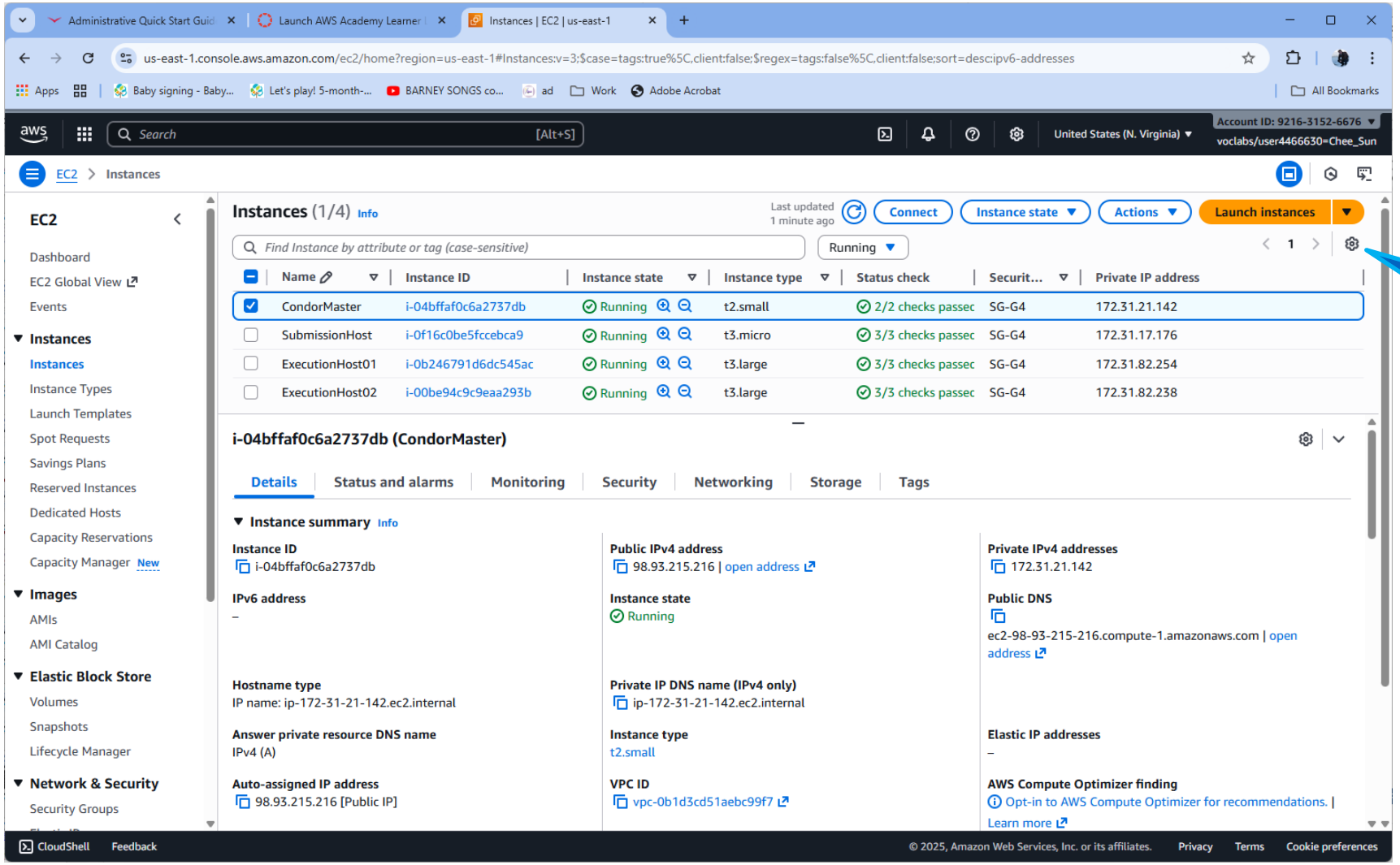
You need to launch 4 instances, following the steps you used in Practical02

3. Select same security group



You can create a new security group, or use back the same security group created in Practical02.
Make sure all of your instances are using the same security group.

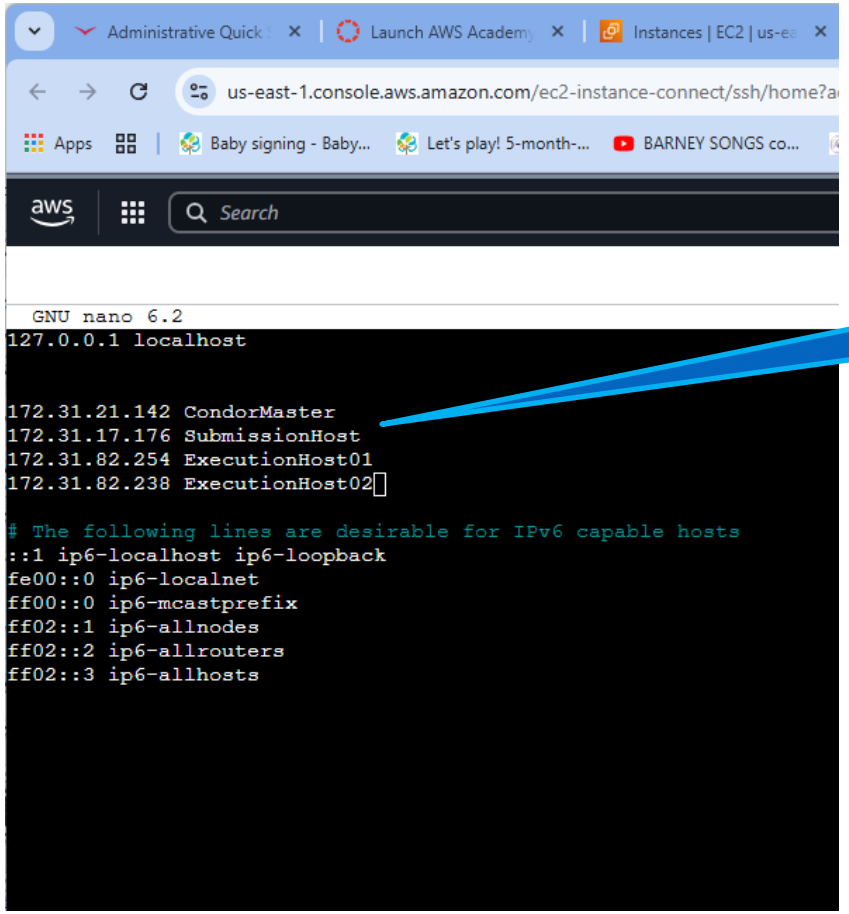
Remark for step 3



You can customize the view in the setting, to show the ip address and security group

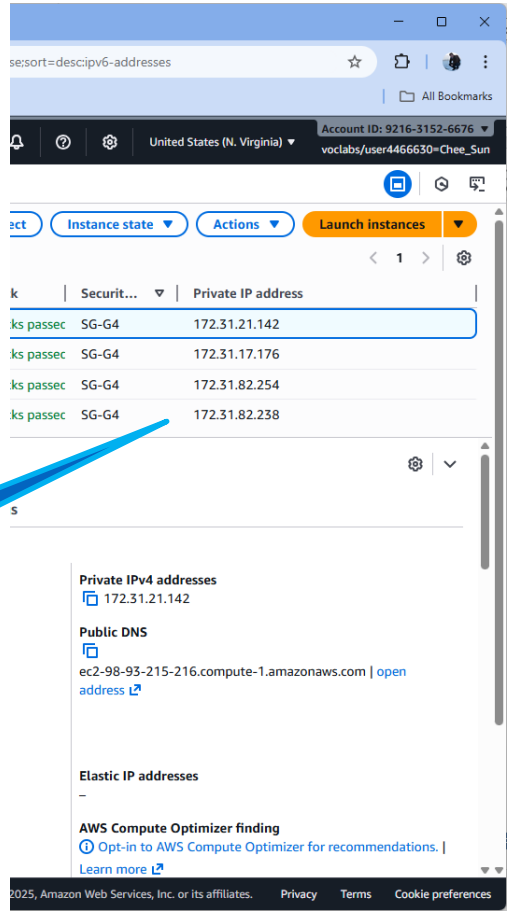
4. edit the /etc/hosts

- Edit the hosts file: sudo nano /etc/hosts

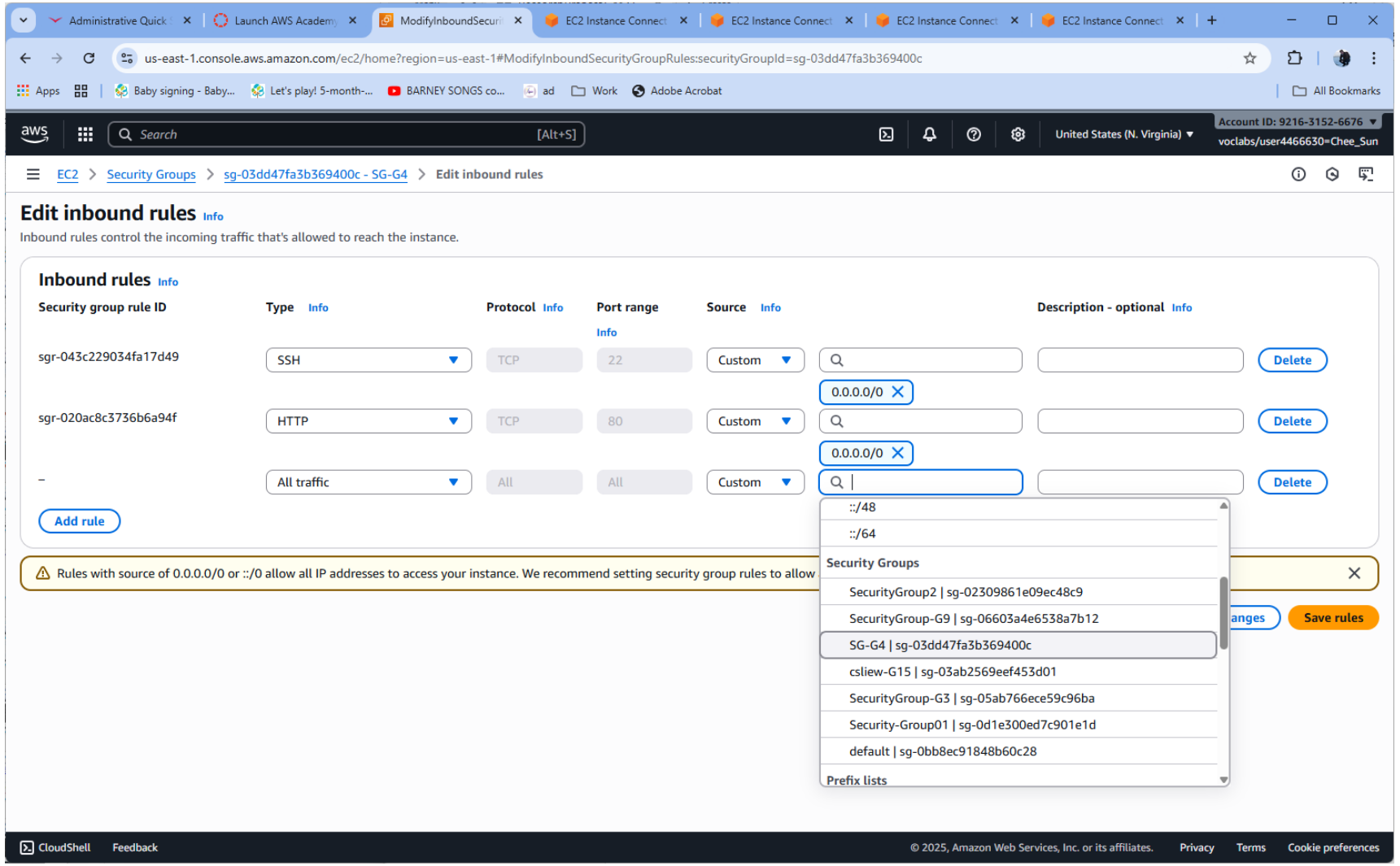


Add 4 lines to define the hosts

Make sure the Ips are matched



5. Configure the security group



Make sure you enable all traffic within your own security group. Otherwise, your machines can't communicate with each other.

6. Install using HTCondor scripts

- For the Central Manager:
  - curl -fsSL https://get.htcondor.org | sudo GET_HTCONDOR_PASSWORD="password" /bin/bash -s -- --no-dry-run --central-manager CondorMaster
- For the Submission Host:
  - curl -fsSL https://get.htcondor.org | sudo GET_HTCONDOR_PASSWORD="password" /bin/bash -s -- --no-dry-run --submit CondorMaster
- For the two Execution Host:
  - curl -fsSL https://get.htcondor.org | sudo GET_HTCONDOR_PASSWORD="password" /bin/bash -s -- --no-dry-run --execute CondorMaster

Remark for step 6

Once you have set up the HTCondor, you can check the daemons running by issuing the systemctl command:
sudo systemctl status condor

Central Manager

```
ubuntu@ip-172-31-21-142:~$ sudo systemctl status condor
● condor.service - Condor Distributed High-Throughput-Computing
     Loaded: loaded (/lib/systemd/system/condor.service; enable
     Active: active (running) since Tue 2025-10-28 01:09:23 UTC
   Main PID: 437 (condor_master)
     Status: "All daemons are responding"
      Tasks: 5 (limit: 4194303)
     Memory: 19.8M
        CPU: 1.567s
     CGroup: /system.slice/condor.service
             ├─437 /usr/sbin/condor_master -f
             ├─573 condor_procd -A /var/run/condo
             ├─575 condor_shared_port
             ├─577 condor_collector
             └─588 condor_negotiator

Oct 28 01:09:23 ip-172-31-21-142 systemd[1]: Star
Oct 28 01:09:23 ip-172-31-21-142 htcondor[519]: N
Oct 28 01:09:23 ip-172-31-21-142 htcondor[522]: N
Oct 28 01:14:31 ip-172-31-21-142 systemd[1]: /lib
Oct 28 01:14:35 ip-172-31-21-142 systemd[1]: /lib
Oct 28 01:14:36 ip-172-31-21-142 systemd[1]: /lib
Oct 28 01:14:38 ip-172-31-21-142 systemd[1]: /lib
Oct 28 01:14:38 ip-172-31-21-142 systemd[1]: /lib
ubuntu@ip-172-31-21-142:~$
```

Submission Host

```
ubuntu@ip-172-31-17-176:~$ sudo systemctl status condor
● condor.service - Condor Distributed High-Throughput-Computing
     Loaded: loaded (/lib/systemd/system/condor.service; enable
     Active: active (running) since Tue 2025-10-28 01:09:18 UTC
   Main PID: 455 (condor_master)
     Status: "All daemons are responding"
      Tasks: 4 (limit: 4194303)
     Memory: 20.4M
        CPU: 967ms
     CGroup: /system.slice/condor.service
             ├─455 /usr/sbin/condor_master -f
             ├─614 condor_procd -A /var/run/condo
             ├─615 condor_shared_port
             └─622 condor_schedd

Oct 28 01:09:18 ip-172-31-17-176 htcondor[488]: N
Oct 28 01:09:18 ip-172-31-17-176 htcondor[543]: C
Oct 28 01:09:18 ip-172-31-17-176 htcondor[546]: N
Oct 28 01:09:18 ip-172-31-17-176 htcondor[556]: N
Oct 28 01:09:18 ip-172-31-17-176 htcondor[559]: C
Oct 28 01:14:25 ip-172-31-17-176 systemd[1]: /lib
Oct 28 01:14:29 ip-172-31-17-176 systemd[1]: /lib
Oct 28 01:14:30 ip-172-31-17-176 systemd[1]: /lib
Oct 28 01:14:32 ip-172-31-17-176 systemd[1]: /lib
Oct 28 01:14:32 ip-172-31-17-176 systemd[1]: /lib
ubuntu@ip-172-31-17-176:~$
```

Execution Host

```
ubuntu@ip-172-31-82-254:~$ sudo systemctl status condor
● condor.service - Condor Distributed High-Throughput-Computing
     Loaded: loaded (/lib/systemd/system/condor.service; enabled
     Active: active (running) since Tue 2025-10-28 01:09:20 UTC;
   Main PID: 461 (condor_master)
     Status: "All daemons are responding"
      Tasks: 4 (limit: 4194303)
     Memory: 63.6M
        CPU: 23.401s
     CGroup: /system.slice/condor.service
             ├─461 /usr/sbin/condor_master -f
             ├─610 condor_procd -A /var/run/condor/procd_pipe -L /var/log/condor/ProcLog
             ├─611 condor_shared_port
             └─615 condor_startd

Oct 28 01:09:20 ip-172-31-82-254 htcondor[545]: Not changing ROOT_MAXKEYS (/proc/sys/kern
Oct 28 01:09:20 ip-172-31-82-254 htcondor[561]: Changing PIPE_USER_PAGES_SOFT (/proc/sys/
Oct 28 01:09:20 ip-172-31-82-254 htcondor[569]: Changing FS_CACHE_DIRTY_BYTES (/proc/sys/
Oct 28 01:09:20 ip-172-31-82-254 htcondor[572]: Changing MAX_RECEIVE_BUFFER (/proc/sys/ne
Oct 28 01:14:29 ip-172-31-82-254 systemd[1]: /lib/systemd/system/condor.service:13: Unkno
Oct 28 01:14:42 ip-172-31-82-254 systemd[1]: /lib/systemd/system/condor.service:13: Unkno
Oct 28 01:14:46 ip-172-31-82-254 systemd[1]: /lib/systemd/system/condor.service:13: Unkno
Oct 28 01:14:47 ip-172-31-82-254 systemd[1]: /lib/systemd/system/condor.service:13: Unkno
Oct 28 01:14:49 ip-172-31-82-254 systemd[1]: /lib/systemd/system/condor.service:13: Unkno
Oct 28 01:14:50 ip-172-31-82-254 systemd[1]: /lib/systemd/system/condor.service:13: Unkno
ubuntu@ip-172-31-82-254:~$
```

Sample output from executing condor_q and condor_status

```
ubuntu@ip-10-135-46-41:~/condor$ condor_q

-- Submitter: ip-10-135-46-41.ap-southeast-1.compute.internal : <10.135.46.41:42992>
46-41.ap-southeast-1.compute.internal
 ID      OWNER            SUBMITTED     RUN_TIME ST PRI SIZE CMD
   2.0   ubuntu          4/17 16:59   0+00:00:00 I  0   0.0  simple 4 10
   2.1   ubuntu          4/17 16:59   0+00:00:00 I  0   0.0  simple 4 11
   2.2   ubuntu          4/17 16:59   0+00:00:00 I  0   0.0  simple 4 12

3 jobs; 0 completed, 0 removed, 3 idle, 0 running, 0 held, 0 suspended

ubuntu@ip-10-135-46-41:~/condor$ condor_status

Name               OpSys      Arch   State     Activity LoadAv Mem   ActvtyTime

slot1@ip-10-135-46 LINUX      X86_64 Claimed   Busy     0.020  3725  0+00:00:04
slot2@ip-10-135-46 LINUX      X86_64 Claimed   Busy     0.000  3725  0+00:00:05
                     Total Owner Claimed Unclaimed Matched Preempting Backfill

        X86_64/LINUX     2     0       2         0       0          0        0

               Total     2     0       2         0       0          0        0
```

On the submission host, you can see the job queue with this command:
condor_q

Once you have set up the HTCondor, you also can see the status of your Condor Pool with this command:
condor_status

7. Let's submit our first Condor Job

- [follow the User's QuickStart Guide at https://htcondor.readthedocs.io/en/latest/users-manual/quick-start-guide.html]
  - Once your Condor pool is up and running, you can try to create a job and submit to your condor pool.
  - You need:
    - A job - the executable/[program that you plan to execute on condor. For instance, a bash script.
    - A job description file - a text file specifying the detail of the job, including the input and output files.

Sample job

- [taken from the User's QuickStart Guide]

```
#!/bin/bash
# file name: sleep.sh

TIMETOWAIT="6"
echo "sleeping for $TIMETOWAIT seconds"
/bin/sleep $TIMETOWAIT
```

Job description file

- [taken from the User's QuickStart Guide]

```
# sleep.sub -- simple sleep job
executable = sleep.sh
log      = sleep.log
output     = sleep.out
error    = sleep.err

should_transfer_files     = Yes
when_to_transfer_output   = ON_EXIT
request_cpus    = 1
request_memory    = 512M
request_disk    = 1G

queue
```

Job description file (continue)

- The first few lines list the input and output files of the job.
- The second part specific the handling of the files, either staging in, or by using shared file system
- The third part shows the specification of the requested resources to run the job. In the example, it requires a single CPU with 512MB RAM.
- The last part of the script is a keyword **queue**, which instruct the Submission Host to submit this script to the Condor Manager.
- Then, you just need to submit this using the command:
  `condor_submit sleep.submit`

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8. Check the log file after execution

```
ubuntu@ip-10-135-46-41:~/condor$ more sleep.log
000 (001.000.000) 04/17 16:56:30 Job submitted from host: <10.135.46.41:42992>
...
001 (001.000.000) 04/17 16:56:31 Job executing on host: <10.135.46.41:48957>
...
006 (001.000.000) 04/17 16:56:35 Image size of job updated: 10
   0  -  MemoryUsage of job (MB)
   0  -  ResidentSetSize of job (KB)
...
005 (001.000.000) 04/17 16:56:35 Job terminated.
   (1) Normal termination (return value 0)
      Usr 0 00:00:00, Sys 0 00:00:00  -  Run Remote Usage
      Usr 0 00:00:00, Sys 0 00:00:00  -  Run Local Usage
      Usr 0 00:00:00, Sys 0 00:00:00  -  Total Remote Usage
      Usr 0 00:00:00, Sys 0 00:00:00  -  Total Local Usage
   56  -  Run Bytes Sent By Job
   8531  -  Run Bytes Received By Job
   56  -  Total Bytes Sent By Job
   8531  -  Total Bytes Received By Job
   Partitionable Resources :    Usage  Request
      Cpus                 :                1
      Disk (KB)            :       17       17
      Memory (MB)          :        0        0
...
```


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THANK YOU

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